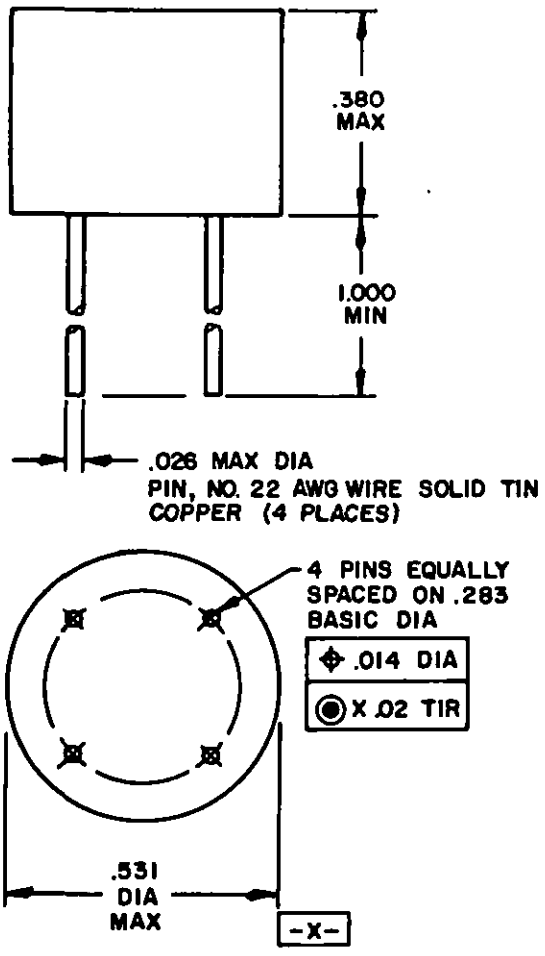
MILITARY SPECIFICATION SHEET

TRANSFORMERS, PULSE, LOW POWER
TYPES TP7RYNNNNH2, TP7RY1100H2 AND TP7RY1200H2

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-21038.

(A) This specification is inactive for new design after the date of this specification.





CIRCUIT DIAGRAM
AND MARKING

INCHES	MM
.014	.36
.02	.5
.026	.66
.283	7.19
.380	9.65
.531	13.49
1.000	25.40

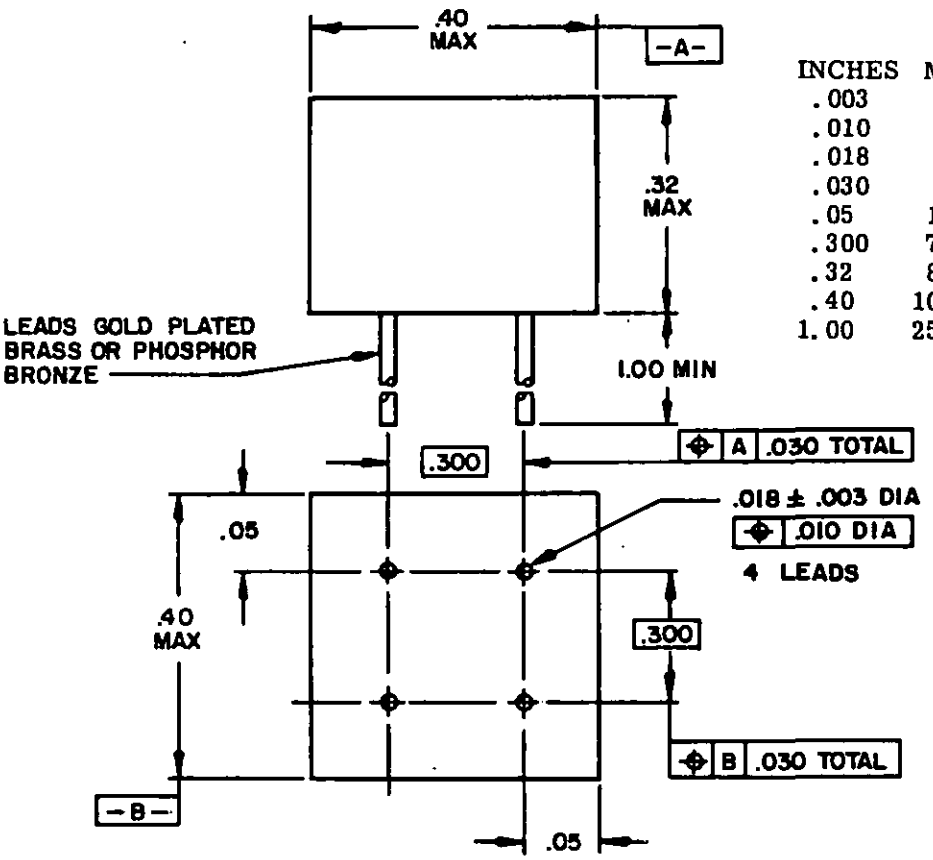
NOTES:
1. *Dimensions are in inches.*
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1.

(A) denotes changes

FSC 5950

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INCHES	MM
.003	.08
.010	.25
.018	.46
.030	.76
.05	1.3
.300	7.62
.32	8.1
.40	10.2
1.00	25.4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2.

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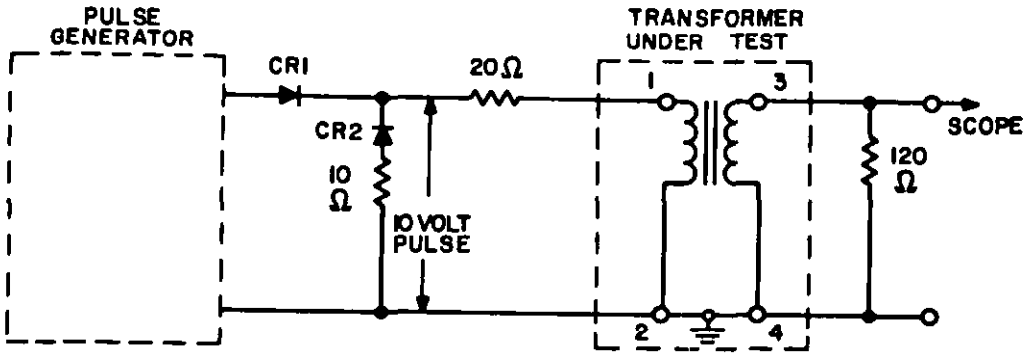
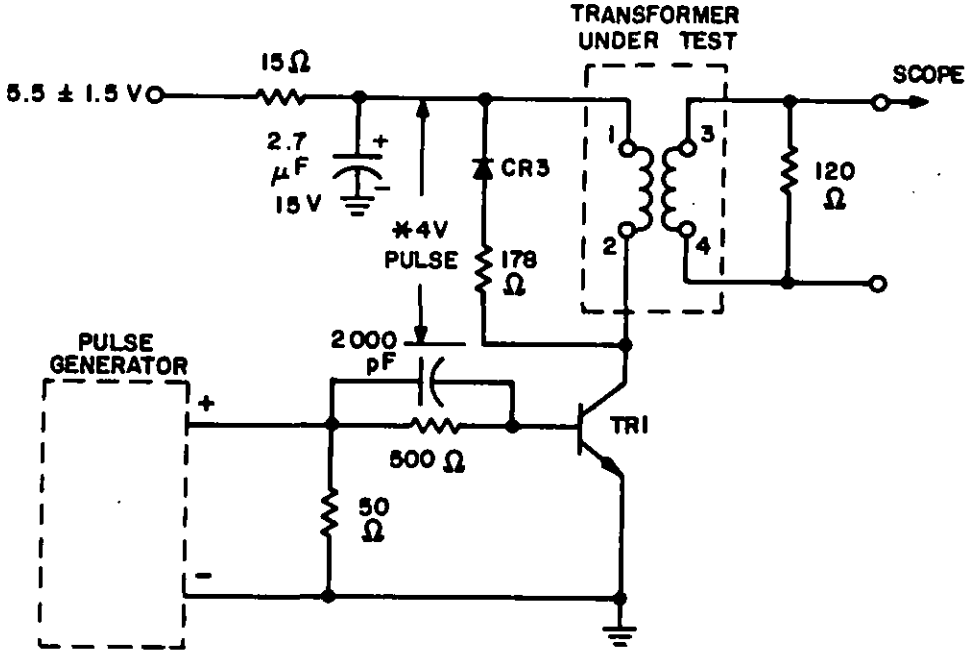


FIGURE 3. Test circuit 1 (for part numbers -001, -002, -003, -005, -006 and -007).



*Use oscilloscope with differential preamplifier; Tektronix type Z or equivalent.

FIGURE 4. Test circuit 2 (for part numbers -004 and -008).

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TABLE I. Electrical characteristics including waveform parameters.

Dash No.	Type desination	Fig. No.	Magnetizing inductance at 115 kHz	Leakage inductance at 3 Mc (μh max)	Input capacitance (pF max)	Coupling capacitance at .95 kHz (pF max)	Input pulse width (μ sec)	Freq. (kHz)	Droop (max)	Output voltage	DC resistance (ohms)		Turns ratio tolerance
											1-2	3-4	
001	TP7RY1100H2	1	6mh ± 20%	10	10	20	2	81.6	.8V	8.0V ±.5V	6Ω (+15% − 5%)	3.1Ω (+15% −25%)	±1%
002	TP7RYNNNNH2	1	6mh ± 20%	10	10	20	2	81.6	.7V	4.7V ±.5V	3.7Ω (+15% − 5%)	2.7Ω (+15% −25%)	±1%
003	TP7RY1200H2	1	6mh ± 20%	10	20	23	4	81.6	3V	9.8V ±.5V	2.82Ω (+15% − 5%)	13.2Ω (+15% −25%)	±1%
004	TP7RYNNNNH2	1	1mh (min. at 250 kHz)	4	40	27	4	81.6	1.3V	9.0V ±.6V	2Ω max	3.0Ω max	±3%
005	TP7RY1100H2	2	6mh ± 20%	10	10	20	2	81.6	.8V	8.0V ±.5V	6Ω (+15% − 5%)	3.1Ω (+15% −25%)	±1%
006	TP7RYNNNNH2	2	6mh ± 20%	10	10	20	2	81.6	.7V	4.7V ±.5V	3.7Ω (+15% − 5%)	2.7Ω (+15% −25%)	±1%
007	TP7RY1200H2	2	6mh ± 20%	10	20	23	4	81.6	3V	9.8V ±.5V	2.82Ω (+15% − 5%)	13.2Ω (+15% −25%)	±1%
008	TP7RYNNNNH2	2	1mh (min. at 250 kHz)	4	40	27	4	81.6	1.3V	9.0V ±.6V	2Ω max	3.6Ω max	±3%

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Case: Encapsulated.

Terminals: Pin type.

Weight: 1.0 ounce.

Terminal strength: MIL-STD-202, method 211, test condition A - pull 5 pounds minimum.

Dielectric withstanding voltage:

At atmospheric pressure: MIL-STD-202, method 301, 100 volts rms for 5 seconds between windings.

Insulation resistance: MIL-STD-202, method 302, test condition B. Not less than 1,000 megohms.

Electrical characteristics: See table I.

Test circuit: See figures 3 and 4.

Life: See MIL-T-21038, life expectancy.

Temperature rise: Not applicable.

Vibration: MIL-STD-202, method 204.

(A) Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Groups A and B tests of MIL-T-21038 shall be applicable.

Part number: M21038/24- (dash number from table I).

Preparing activity:
Air Force - 85

(Project 5950-0587-16)

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